



Digihaul Take Home Test – Data Analyst

Disclaimer

This document is prepared for Digihaul recruitment assessments. All datasets and instructions herein are confidential and intended solely for candidate evaluation.



Overview

You are provided with a logistics dataset including:

- shipments
- pickup_locations
- delivery_locations
- trips
- drivers
- vehicles
- shippers
- carriers
- shipment_status

Your goal is to build a **production-ready Power BI report** with:

- Robust **DAX measures**
 - Strong **data modeling**
 - Clear **storytelling aligned with Digihaul's operational KPIs and Additional point for any other findings.**
-



Inferred Data Relationships (ERD)

Fact Table:

- **shipments**
PK: **ShipmentID**
FK: **ShipperID**, **CarrierID**, **StatusID**

Dimensions:

- **shippers** — **ShipperID**
- **carriers** — **CarrierID**
- **shipment_status** — **StatusID**
- **vehicles** — **VehicleID**, FK: **CarrierID**
- **drivers** — **DriverID**, FK: **CarrierID**

Supporting Tables:

- **trips** — FK: **ShipmentID**, **VehicleID**, **DriverID**
- **pickup_locations** — FK: **ShipmentID**, **StatusID**
- **delivery_locations** — FK: **ShipmentID**, **StatusID**



Power BI Dashboard (Core Task)



Data Modelling:

- Create relationships per ERD
- Apply star schema principles
- Format data types and use friendly column names



DAX Measures:

- On-Time Pickup Rate (%)
- On-Time Delivery Rate (%)
- Average Pickup Delay (minutes)
- Average Delivery Delay (minutes)
- Pickup Time Window Compliance %
- Delivery Time Window Compliance %
- Average Shipment Lead Time (Created → Completed)

- Total Shipment Count (by status, region, carrier)
- Average Distance per Vehicle Type
- Driver KPIs: Number of Trips, On-Time Rate, Average Delay
- **Advanced DAX measures:**
 - Use of **iterator functions** (e.g., `COUNTX`, `SUMX`, `AVERAGEX`) for row context calculations
 - **Time intelligence** functions (e.g., MTD, YTD, MoM) for trend analysis
 - **USERELATIONSHIP** to activate inactive relationships where necessary (e.g., for alternate date fields)

Visualizations:

- **Data visualization techniques:**
 - Drill-through pages for shipment and driver details
 - Page tooltips to provide contextual summaries
 - Page navigation buttons for intuitive report flow
 - Synced slicers for consistent filtering across pages
 - Shipment Status Funnel
 - Pickup & Delivery Compliance Gauges
 - Carrier Performance Leaderboard
 - Shipper-wise On-Time Delivery Bar Chart
 - Vehicle Utilization Summary
 - Driver On-Time Performance Leaderboard
 - (Optional) Any additional insights/visualizations you recommend [Bonus Points]
-

SQL Stretch Tasks

1. Retrieve shipments with **multiple pickups and multiple deliveries**
2. Calculate **average pickup delay** grouped by **region**
3. Identify **vehicles with utilization < 50%** last month

4. List **top 5 carriers with highest average delivery delays**

Deliverables

-  Power BI `.pbix` file (with DAX, relationships, layout)
-  `.sql` script with solutions to the SQL tasks
-  (Optional) `README.md` summarising:
 - Your approach
 - Any assumptions made
 - Recommendations or insights

Note: zip all the deliverables in a single folder with your name as folder name.

Submission Deadline

 **Submit within 7 Days** from the time you receive this task.
